

THE EXTREME TAU TEST FOR SINGLE LAG ECONOMETRIC MODELS

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SUMMARY

Although distributed lag models play an increasingly important role in econometrics, statistical tests for the existence of such models are often inadequate or non-existent. Here such a non-parametric test is developed for significant co-movement at an *a priori* unknown lag between time series, each consisting of mutually independent observations.

1. Test Statistic

Consider two series $Y_1, Y_2, \dots, Y_{n+m}$; $X_1, X_2, \dots, X_{n+m}$ of mutually independent observations. We wish to test the null hypothesis

H_0 : The X-series and Y-series are independent,

against the alternative

H_a : There exists a $k \geq 0$ such that co-movement between X_i and Y_{i+k} is significant.

If k is known *a priori* then Kendall's well-known rank correlation coefficient

$$\sum_{i=1}^N \sum_{j=1}^N \text{sign}(Y_{i+k} - Y_{j+k}) \text{sign}(X_i - X_j) / (N(N-1))$$

where

$$N = n + m - k$$

can be used. However, most often, no single lag, k , is known in advance. More readily it will be possible to set a maximum, m , for k such that significant co-movement must be tested for $k = 0, 1, \dots, m$. Let

$$\tau_k^{(n)} \equiv \sum_{i=1}^n \sum_{j=1}^n \text{sign}(Y_{i+k} - Y_{j+k}) \text{sign}(X_i - X_j) / n(n-1)$$

then

$$\tau^{(n)} \equiv \max_{k=0, \dots, m} (|\tau_k^{(n)}|)$$

is a possible test statistic for testing H_0 against H_a .

Note that $\tau_k^{(n)}$ does not utilise all the available information, for example $\tau_0^{(n)}$ is based on the pairs $(X_1, Y_1), \dots, (X_n, Y_n)$, thus ignoring the last m pairs. The reasons for defining $\tau_k^{(n)}$ thus are:

- (i) intuitively it can be argued that each lag must have an "equal chance" to compete as candidate for the single lag model;
- (ii) this formulation of $\tau_k^{(n)}$ is mathematically more tractable;
- (iii) also if $n \gg m$ the influence of pairs not utilised is small.

2. Asymptotic Distribution of $\tau^{(n)}$ under H_0

Let

$$t_k(i, j) \equiv \text{sign}(Y_{i+k} - Y_{j+k}) \text{sign}(X_i - X_j)$$

$$L_k^{(n)} \equiv \sum_{i=1}^n \sum_{j=1}^n (E[t_k(i, j)|X_i, Y_{i+k}] + E[t_k(i, j)|X_j, Y_{j+k}] - E[t_k(i, j)]).$$

From Hoeffding's theory of U-statistics it readily follows that:

Lemma 1

$$(n)^{1/2} \tau_k^{(n)} - (n)^{1/2} L_k^{(n)} / (n(n-1)) \rightarrow 0 \text{ in probability under } H_0.$$

Theorem 1

$(n)^{1/2} L_k^{(n)} / (n(n-1))$ and $(n)^{1/2} L_\ell^{(n)} / (n(n-1))$ are asymptotically independent normal variates for $0 \leq k < \ell \leq m$ under H_0 .

Proof:

$$\text{Let } F(x, y) \equiv P[X \leq x, Y \leq y],$$

$$\bar{F}(x, y) \equiv P[X \geq x, Y \geq y]$$

and

$$\phi(X_i, Y_{i+k}) \equiv F(X_i, Y_{i+k}) + \bar{F}(X_i, Y_{i+k}) - \frac{1}{2}$$

Then

$$E[t_k(i,j)|X_i, Y_{i+k}] = 2\phi(X_i, Y_{i+k})$$

and

$$E[t_k(i,j)|X_j, Y_{j+k}] = 2\phi(X_j, Y_{j+k}).$$

Hence

$$\begin{aligned} L_k^{(n)} &= \sum_{i \neq j} [2\phi(X_i, Y_{i+k}) + 2\phi(X_j, Y_{j+k}) - (4E[F(X, Y)] - 1)] \\ &= 4(n-1) \sum_{i=1}^n \phi(X_i, Y_{i+k}). \end{aligned}$$

The sequence

$$(\phi(X_i, Y_{i+k}), \phi(X_i, Y_{i+\ell})), \quad i = 1, \dots, n$$

is a sequence $(\ell-k)$ -dependent random vectors and from a result by Hoeffding and Robbins (1948) follows that

$$(n)^{-1/2} (\sum \phi(X_i, Y_{i+k}), \sum \phi(X_i, Y_{i+\ell}))$$

is asymptotically normally distributed with mean $(0,0)$ and covariance matrix

$$\begin{bmatrix} A & B \\ B & C \end{bmatrix}$$

where

$$A \equiv E[\phi(X_1, Y_{1+k})^2] + \sum_{j=2}^{\ell-k+1} E[\phi(X_1, Y_{1+k})\phi(X_j, Y_{j+k})]$$

$$= \frac{1}{36}$$

$$C = \frac{1}{36}$$

$$B \equiv E[\phi(X_1, Y_{1+k})\phi(X_1, Y_{1+\ell})] + \sum_{j=2}^{\ell-k+1} E[\phi(X_1, Y_{1+k})\phi(X_j, Y_{j+\ell})]$$

$$+ \sum_{j=2}^{\ell-k+1} E[\phi(X_j, Y_{j+k})\phi(X_1, Y_{1+\ell})]$$

$$= 0.$$

Hence

$(n)^{1/2}L_k^{(n)} / (n(n-1))$ and $(n)^{1/2}L_q^{(n)} / (n(n-1))$ are asymptotically independent normal variates with mean zero and variance

$$\sigma^2 = \frac{4}{9}.$$

Theorem 2

If $X_1, \dots, X_{n+m}$ and $Y_1, \dots, Y_{n+m}$ are two series of mutually independent identically distributed random variates, independent of each other, then $\tau_0^{(n)}, \dots, \tau_m^{(n)}$ are asymptotically independent identically distributed normal variates with mean zero and variance $4/9n$.

Proof:

The result follows immediately from Lemma 1 and Theorem 1.

Remark

Through tedious calculations it can be shown that the exact correlation between $\tau_k^{(n)}$ and $\tau_{k+1}^{(n)}$ is given by

$$\rho(n) = (-6(2n-3) - 2(n-2)(n-3)) / (n(n-1)(2n+5))$$

3. Application of the Extreme Tau Test

Six series of 100 random numbers each (three uniformly distributed between 0 and 100, and three $N(0,1)$ distributed) were compared and the extreme tau test applied over lags 0 to 10. Of the thirty extreme tau-values thus obtained ($\max(\max \tau, \min \tau)$) only one was significant which is in accordance with the expected value of 1.5. In contrast, applying the t-test without allowing for the fact that the statistic used is actually the extreme of eleven t-statistics, the type I error occurred thirteen times.

4. Tables for practical use

It has been shown that for a large number of observations the distribution of $\tau^{(n)}$ is, under H_0 , the distribution of the extreme value of $(m+1)$ independent identically distributed normal variates with mean zero and variance $4/9n$. Critical values for this statistic are well tabulated. See Pearson and Hartley (1966).

For n small and $m < n$ tables are attached. However to obtain exact frequencies it is necessary to match $n!$ permutations of ranks $1, \dots, n$ to $(n+m)!$ permutations of ranks $1, \dots, n+m$ and each time calculate and compare the resulting $(m+1)$ tau-values. This soon becomes prohibitive. The remark at the end of paragraph 3 indicates that $\rho(n)$ is quite small even for n relatively small. This suggests a good transient approximation may be possible using the exact distribution for a single tau but treating the $\tau_k^{(n)}$ as independent

of each other. Comparison shows this approximation to be indeed very good even for n as small as 5. See Grobbelaar (1974). Hence in the tables below the exact distributions are used for $n = 3, 4, 5$ whilst the transient approximation is used for $n = 6, \dots, 10$. As in the case of the single tau the normal approximation was found acceptable for $n > 10$. The tables are only for maximum tau as the distribution for minimum tau is simply the mirror image

$$P[\tau_{\min} \geq -a] = P[\tau_{\max} \leq a].$$

For $n = 3, 4$ and $m < n$ the full distribution is given whilst for $n > 4$ the tables are sufficiently complete to allow one-tailed testing on 10% level.

TABLE 1

$P[\sum_{k=0,1,\dots,m-1}^{\max} S_k = s]$ for $n = 3, 4$; s and m as given, where $S_k \equiv \#(t_k(i,j) = 1 | i < j)$

s	n=3 m=2	n=4 m=2	n=4 m=3
6		0,08264	0,12153
5		0,23958	0,32726
4		0,32848	0,35590
3	0,31944	0,25000	0,16088
2	0,48611	0,08819	0,03009
1	0,18056	0,01042	0,00405
0	0,01389	0,00069	0,00029

TABLE 2

$P[\sum_{k=0,1,\dots,m-1}^{\max} S_k \leq s]$ for n, m and s as given, S_k as described above

n = 5			
m			
s	2	3	4
9	0,98336	0,97514	0,96723
8	0,91709	0,87749	0,84235
7	0,77288	0,67510	0,60486

TABLE 2 (Continued)

 $n = 6$

$\begin{matrix} m \\ s \end{matrix}$	2	3	4	5
14	0,99722	0,99584	0,99446	0,99307
13	0,98340	0,97521	0,96708	0,95902
12	0,94522	0,91896	0,89343	0,86862
11	0,86852	0,80941	0,75433	0,70299
10	0,74630	0,64472	0,55697	0,48116

 $n = 7$

$\begin{matrix} m \\ s \end{matrix}$	2	3	4	5	6
20	0,99960	0,99940	0,99921	0,99901	0,99881
19	0,99722	0,99584	0,99446	0,99307	0,99170
18	0,98931	0,98401	0,97874	0,97350	0,96828
17	0,97007	0,95544	0,94103	0,92684	0,91287
16	0,93214	0,89996	0,86889	0,83890	0,80993
15	0,86852	0,80941	0,75433	0,70299	0,65515

 $n = 8$

$\begin{matrix} m \\ s \end{matrix}$	2	3	4	5	6	7
27	0,99995	0,99993	0,99990	0,99988	0,99985	0,99983
26	0,99960	0,99940	0,99901	0,99901	0,99881	0,99861
25	0,99826	0,99740	0,99653	0,99567	0,99480	0,99394
24	0,99450	0,99176	0,98903	0,98631	0,98360	0,98089
23	0,98591	0,97894	0,97202	0,96515	0,95833	0,95156
22	0,96909	0,95400	0,93914	0,92451	0,91011	0,89594
21	0,93992	0,91125	0,88345	0,85650	0,83037	0,80504
20	0,89427	0,84568	0,79972	0,75627	0,71517	0,67631

TABLE 2 (Continued)

		n = 9						
s	m	2	3	4	5	6	7	8
35		0,99999	0,99999	0,99999	0,99999	0,99998	0,99998	0,99998
34		0,99995	0,99993	0,99990	0,99988	0,99985	0,99983	0,99980
33		0,99976	0,99964	0,99952	0,99939	0,99927	0,99915	0,99903
32		0,99915	0,99872	0,99829	0,99787	0,99744	0,99701	0,99659
31		0,99758	0,99637	0,99516	0,99395	0,99275	0,99154	0,99034
30		0,99412	0,99120	0,98828	0,98537	0,98247	0,97958	0,97670
29		0,98737	0,98112	0,97491	0,96874	0,96260	0,95650	0,95045
28		0,97541	0,96335	0,95143	0,93966	0,92804	0,91655	0,90522
27		0,95588	0,93456	0,91371	0,89333	0,87340	0,85382	0,83487
26		0,92624	0,89142	0,85791	0,82567	0,79463	0,76476	0,73602
25		0,88413	0,83133	0,78168	0,73500	0,69111	0,64983	0,61103

n = 10

s	m	2	3	4	5	6	7	8	9
45		1,00000	1,00000	1,00000	1,00000	1,00000	1,00000	1,00000	1,00000
44		0,99999	0,99999	0,99999	0,99999	0,99998	0,99998	0,99998	0,99998
43		0,99997	0,99996	0,99994	0,99993	0,99991	0,99990	0,99988	0,99987
42		0,99988	0,99983	0,99977	0,99971	0,99965	0,99960	0,99954	0,99948
41		0,99964	0,99946	0,99928	0,99911	0,99893	0,99875	0,99857	0,99829
40		0,99905	0,99858	0,99811	0,99764	0,99716	0,99669	0,99622	0,99575
39		0,99779	0,99668	0,99558	0,99448	0,99338	0,99228	0,99118	0,99009
38		0,99532	0,99299	0,99066	0,98834	0,98602	0,98371	0,98141	0,97911
37		0,99087	0,98634	0,98183	0,97734	0,97287	0,96842	0,96399	0,95958
36		0,98340	0,97521	0,96708	0,95902	0,95103	0,94311	0,93525	0,92745
35		0,97160	0,95770	0,94400	0,93049	0,91718	0,90406	0,89113	0,87838
34		0,95392	0,93168	0,90996	0,88875	0,86803	0,84780	0,82804	0,80873
33		0,92877	0,89507	0,86261	0,83131	0,80116	0,77210	0,74409	0,71710
32		0,89462	0,84617	0,80034	0,75700	0,71600	0,67723	0,64055	0,60586

REFERENCES

- GROBBELAAR, J.W. (1974). *Die ko-beweging van tydreekse en geassosieerde verdelings-vrye toetse*. Unpublished Ph.D. thesis, University of South Africa.
- HOEFFDING, W. and ROBBINS, H. (1948). The central limit theorem for dependent random variables. *Duke Math. J.*, 18 773-780.
- PEARSON, E.S. and HARTLEY, H.O. (Ed.) (1966). *Biometrika tables for statisticians* 1 (3rd Ed.). Cambridge University Press.

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